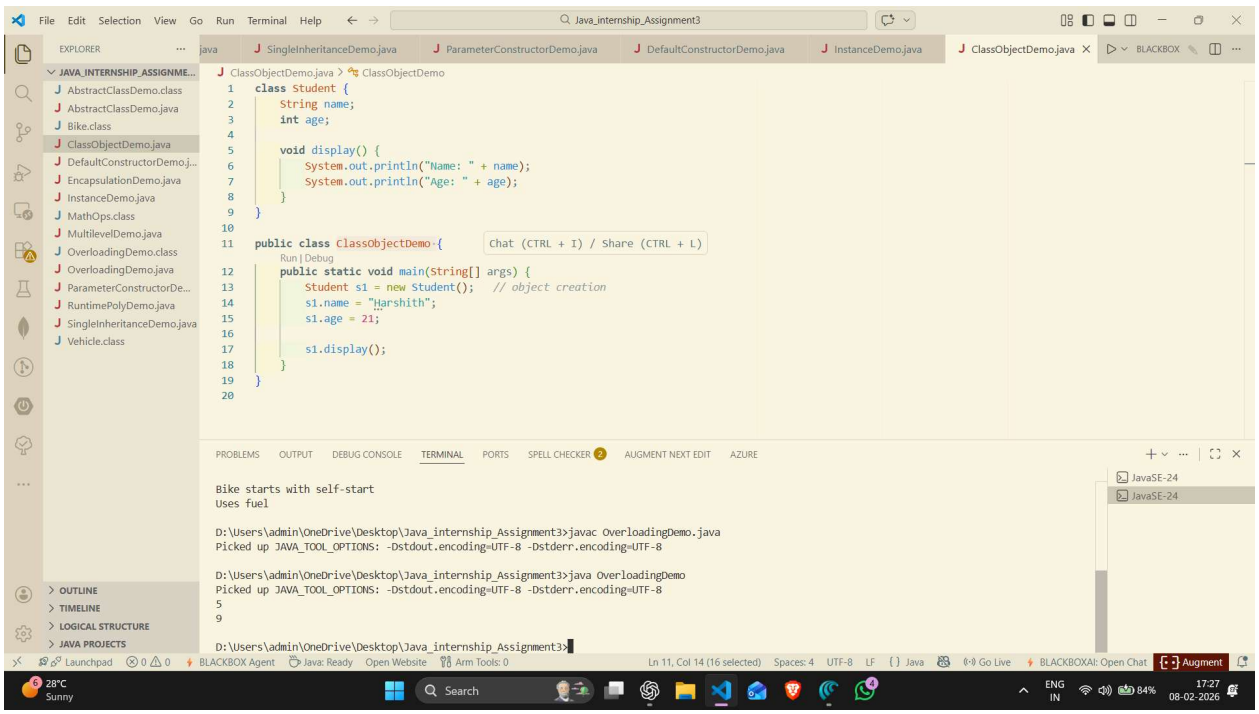
HARSHITH V-1MJ22CS073.(DATE:(07-02-2026 & 08-02-2026))

Java Programming Internship with progarm and outputs: Assignment 3

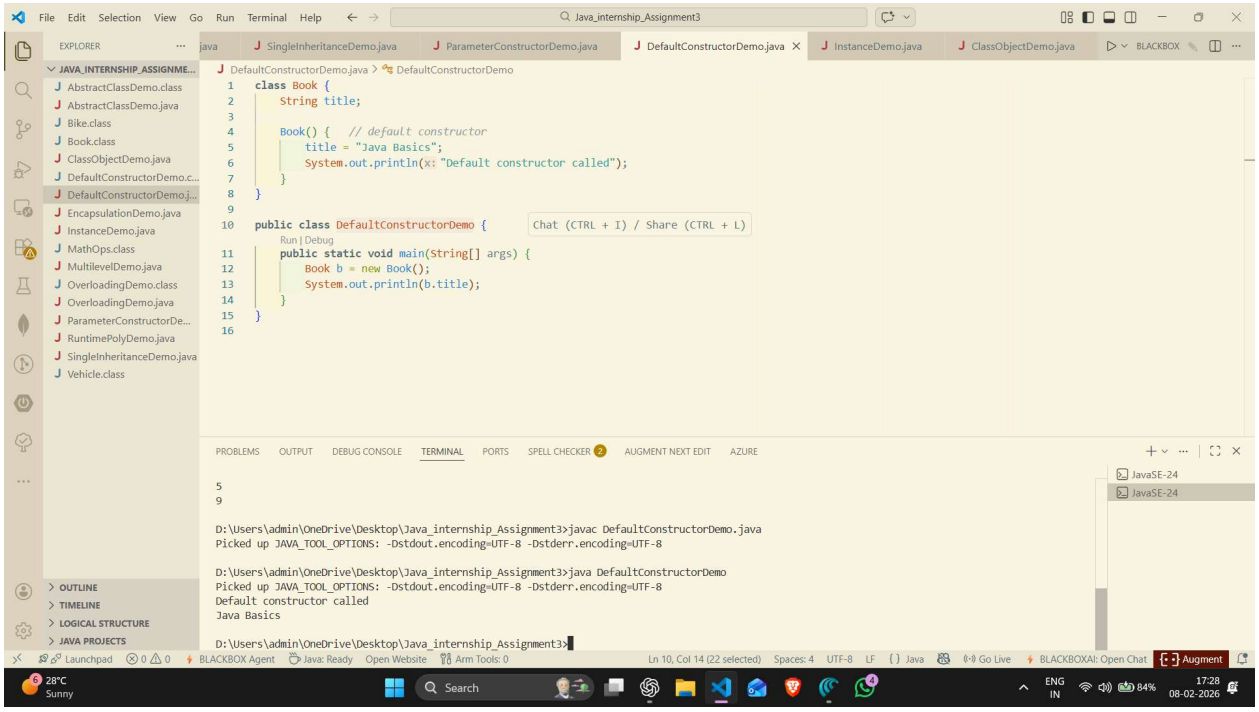
1. Class and Object Demo.

Concept: Demonstrates the basic blueprint of a class and how to instantiate an object to access its members.



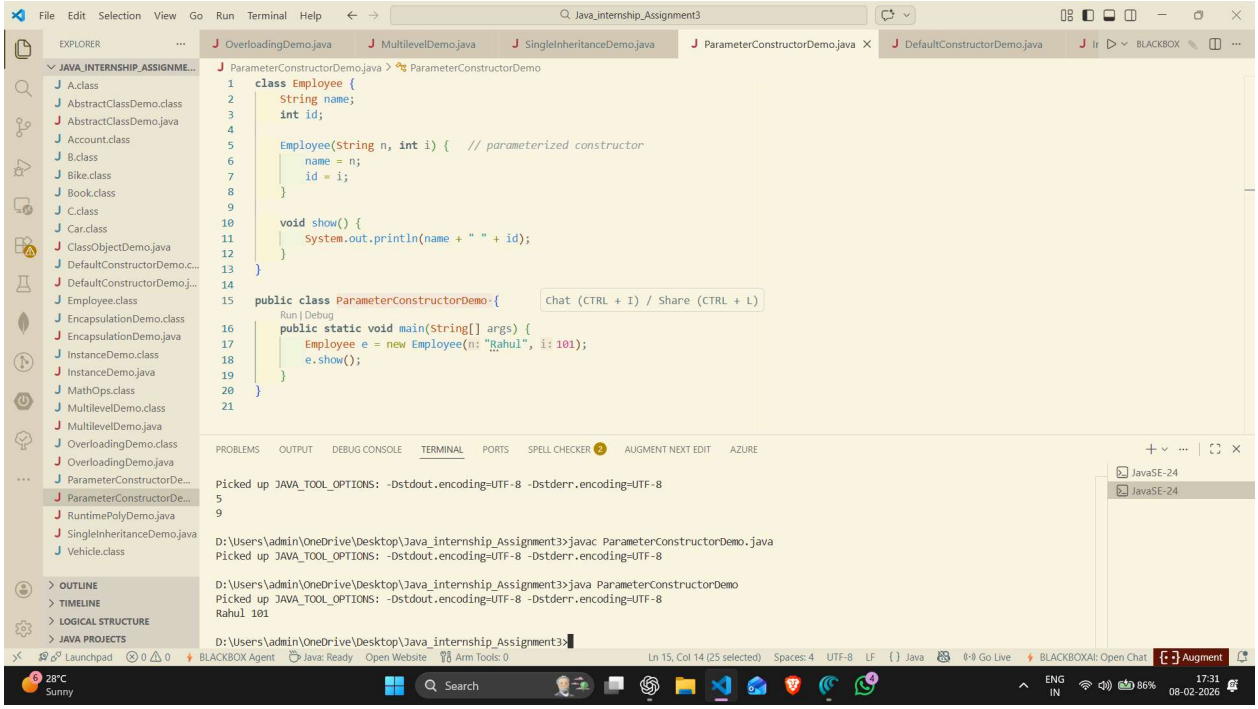
2. Default Constructor

Concept: Shows how a default (no-argument) constructor initializes object state at the time of creation.



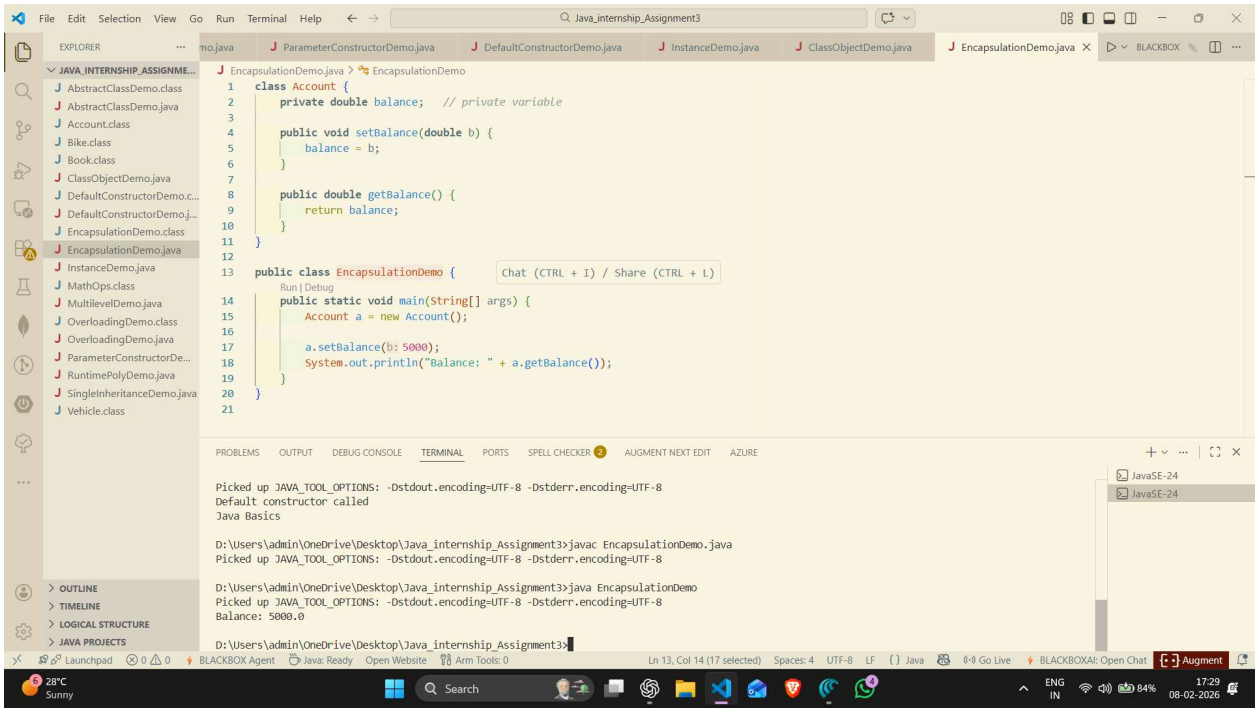
3. Parameterized Constructor

Concept: Demonstrates passing values to a constructor to initialize an object with specific data.



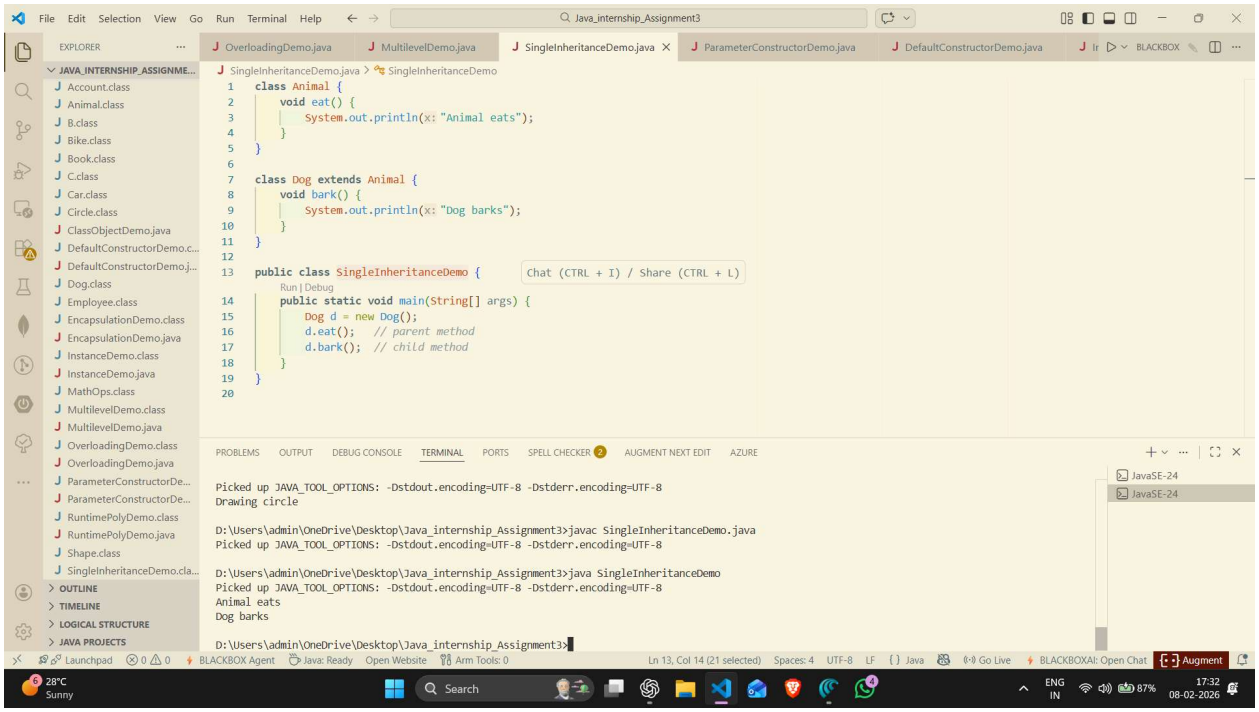
4. Encapsulation

Concept: Illustrates data hiding by using private variables and providing access through public getter and setter methods.



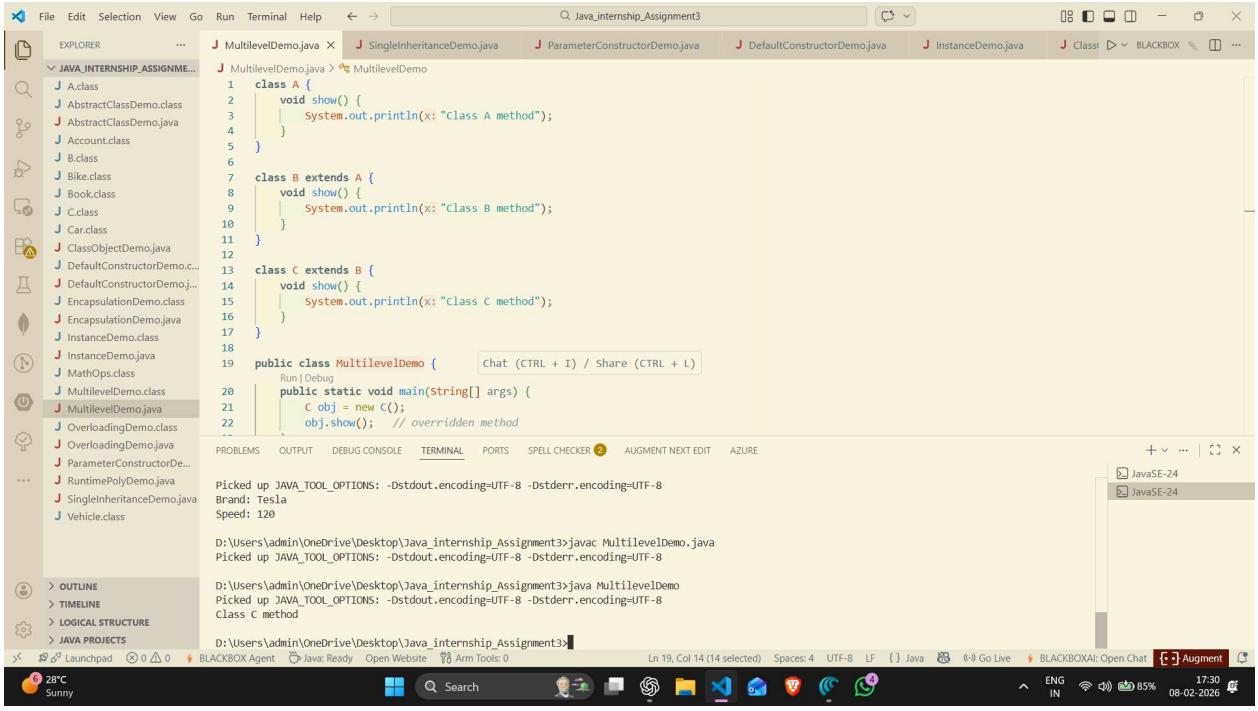
5. Single Inheritance

Concept: A child class (Dog) inheriting the properties and methods of a single parent class (Animal).



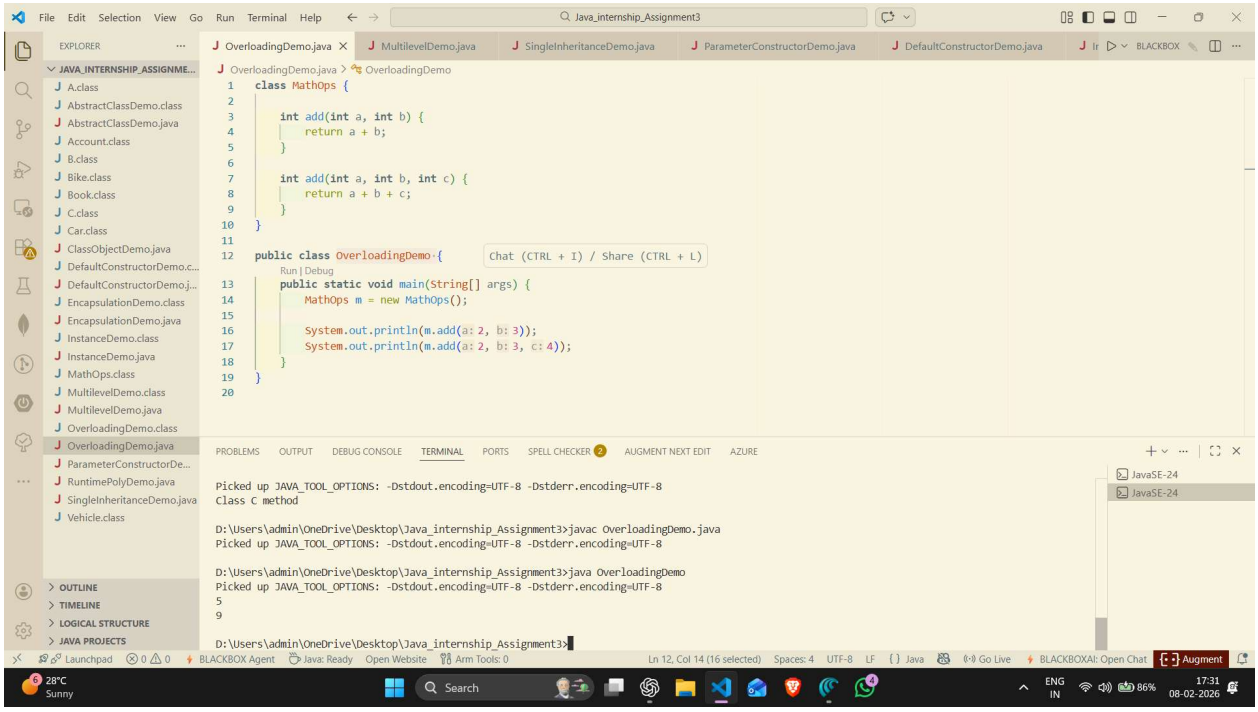
6. Multilevel Inheritance

Concept: A derived class inheriting from another derived class, forming a chain (A -> B -> C).



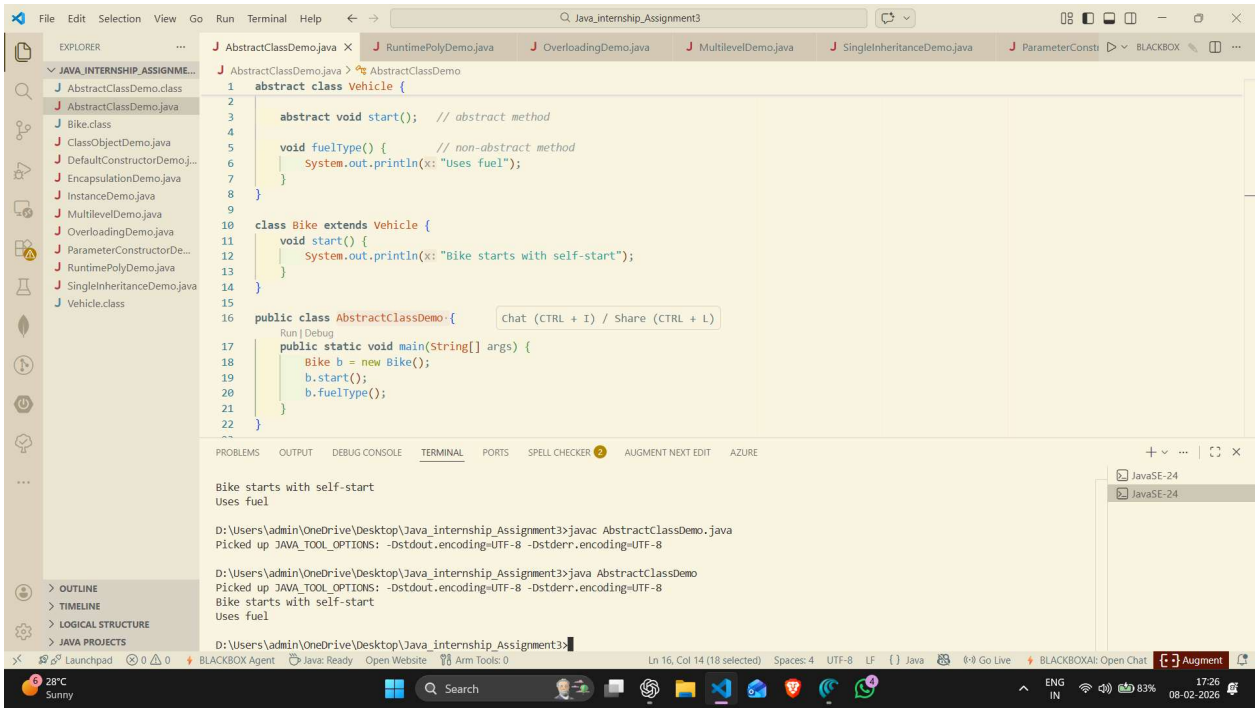
7. Method Overloading (Compile-time Polymorphism)

Concept: Defining multiple methods with the same name but different parameter lists within the same class.



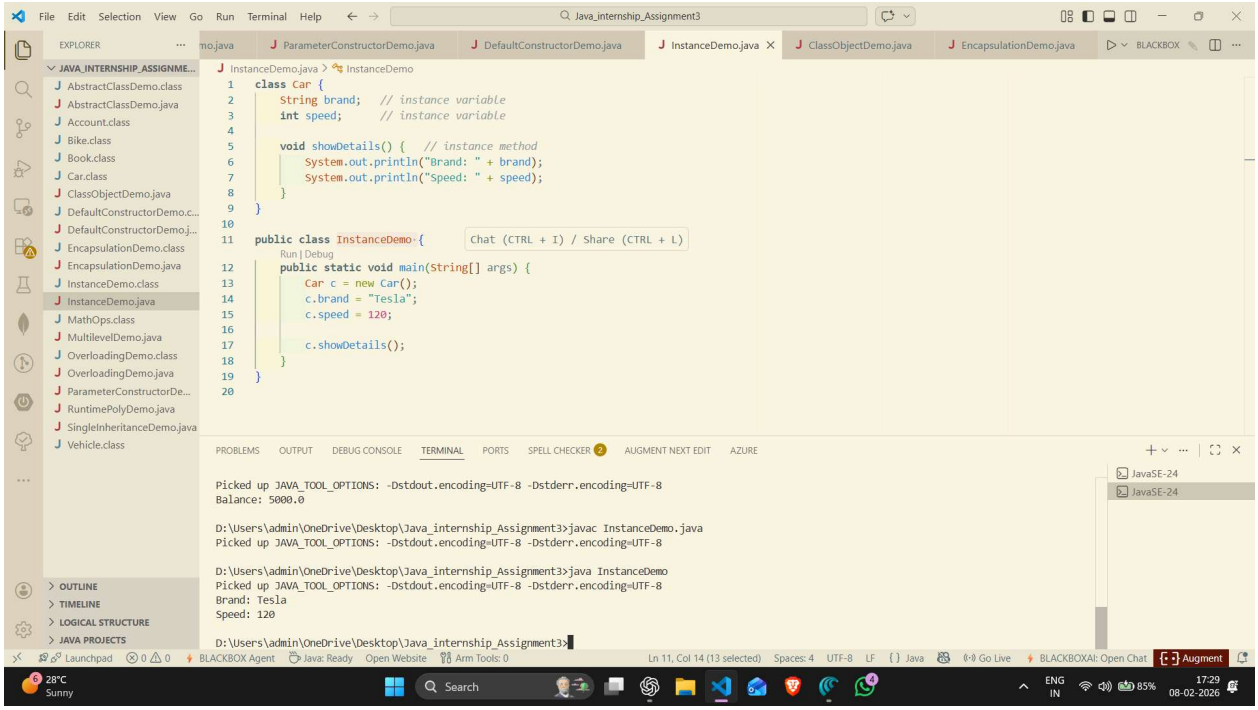
8. Abstract Class

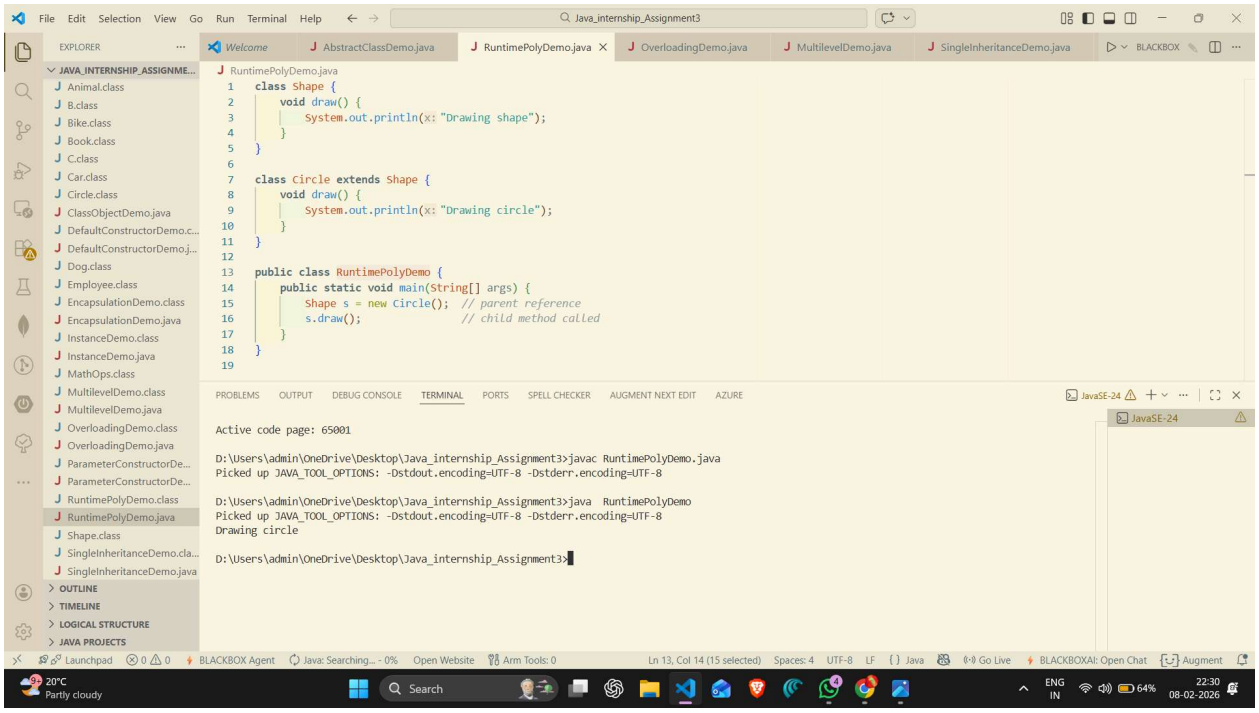
Concept: A restricted class that cannot be used to create objects; it may contain abstract methods (without a body) and regular methods.



9. Instance Variables (State Management)

Concept: Demonstrates how instance variables (like `brand` and `speed`) store the unique state for each object of a class.



10.Runtime Polymorphism (Method Overriding)

Concept: Runtime polymorphism, also known as **Dynamic Method Dispatch**, occurs when a call to an overridden method is resolved at runtime rather than compile-time. This is achieved through **upcasting**, where a parent class reference variable points to a child class object.